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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – FLOWCHART-TO-CODE CONVERSION

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 2 – Flowchart-To-Code Conversion
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	J.Meher Koushik	**Reg. No.:**	192525211
Section / Batch:	AI&ML	**Date:**	04/09/2026

Code of Conduct

I, J.Meher Koushik (192525211) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: J.Meher Koushik

Instructions

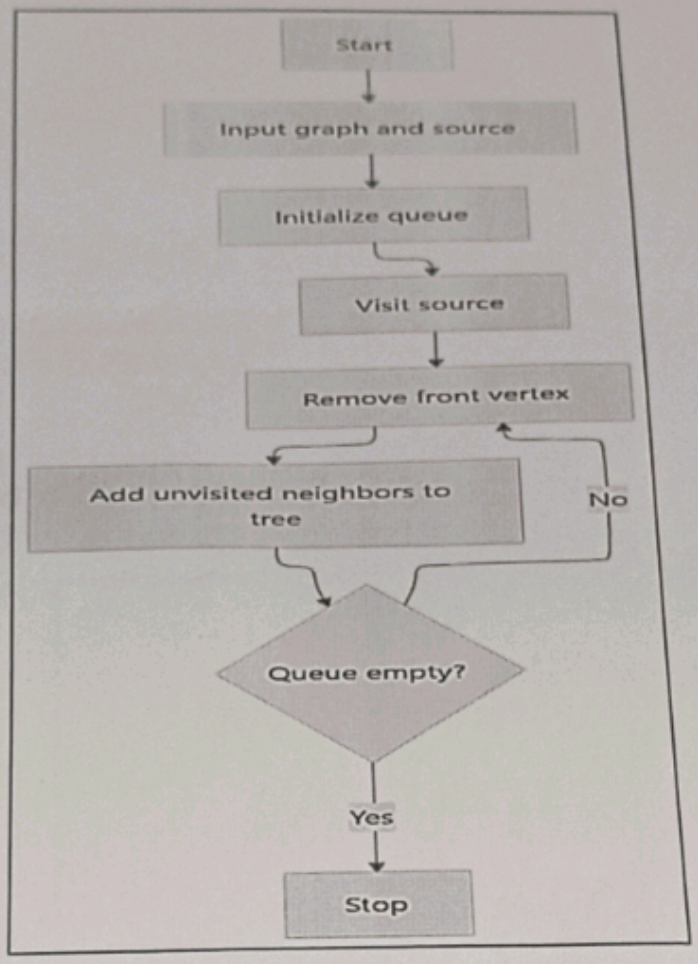
- This test contains 5 Flowchart-to-code conversion questions. Total: 100 Marks.
- When a student answer using Flowchart-to-code conversion should evaluate based on the ability to design clear, logical, and efficient code solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Graph Traversal	BFS	1	20
Shortest Path Algorithm	Dijkstra's Algorithm	2	20
Shortest Path Algorithm	Unweighted Graph	3	20
Graph Traversal	BFS	4	20
Minimum Spanning Tree	Prim's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

Annexure A – Pseudocode Writing Task

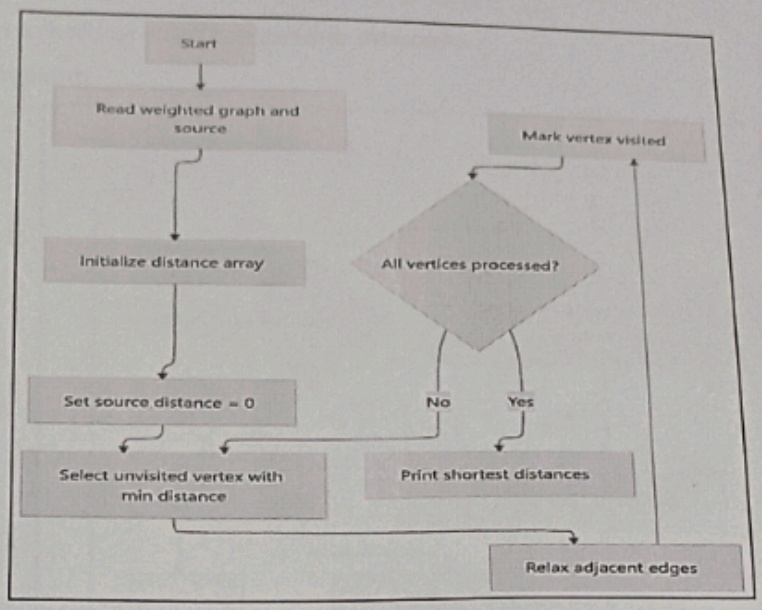
1. Convert the flowchart for generating a BFS tree into code. (20 Marks)

Flowchart Diagram:



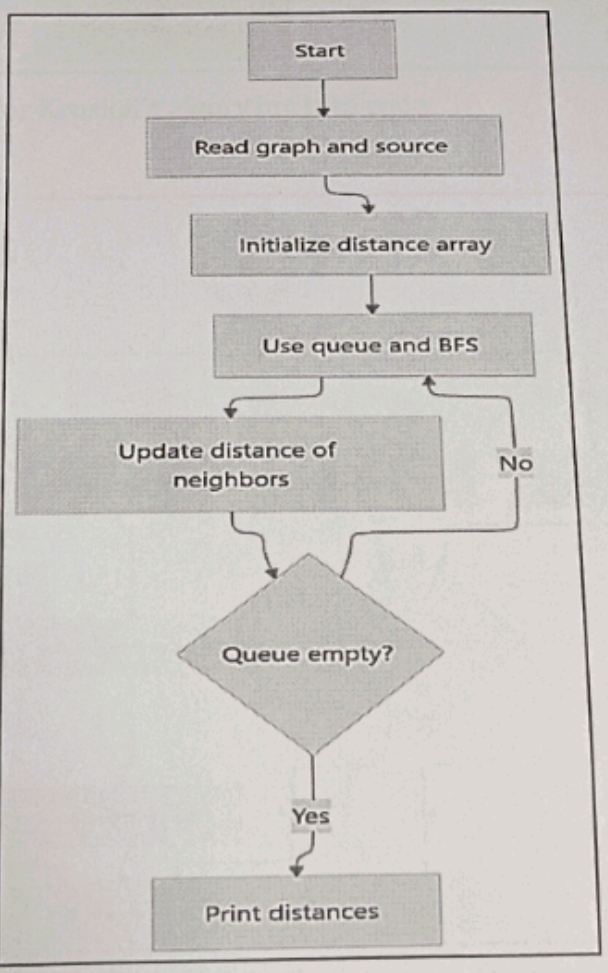
2. Convert the flowchart for Dijkstra's shortest path algorithm into code. (2

Flowchart Diagram:



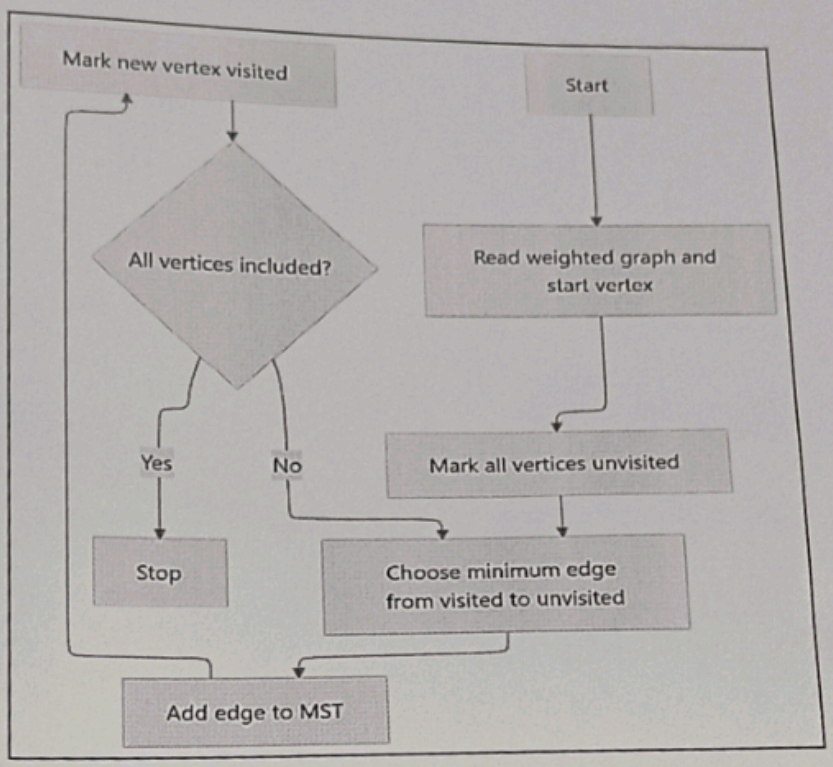
3. Convert the flowchart for shortest path in an unweighted graph into code. (20 Marks)

Flowchart Diagram:



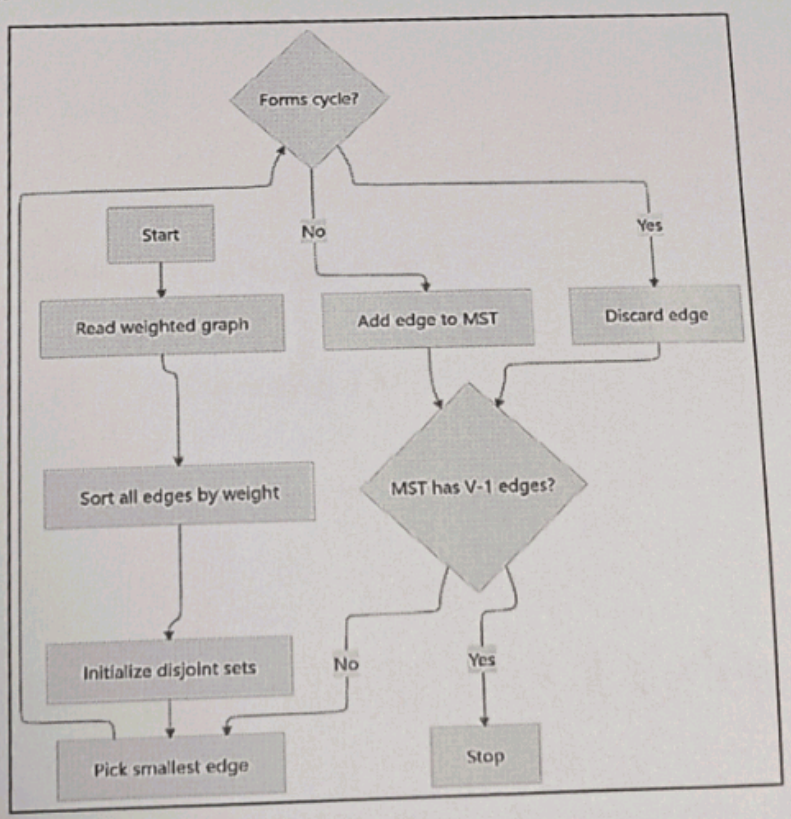
4. Convert the flowchart for Prim's algorithm into code. (20 M

Flowchart Diagram:



5. Convert the flowchart for Kruskal's algorithm into code.

Flowchart Diagram:



S TREE

```
#include <stdio.h>
#define Max 20
int graph[Max][Max];
int visited[MAX];
int queue[MAX];
int front=0, rear=0;
int n;

void BFS(int start)
{
  visited[start]=1;
  queue[rear++]=start;
 printf("BFS Traversal:");
while(front<rear):
 {
  int current=queue[front++];
  printf("%d", current);
  for(int i=0;i<n;i++)
  {
    if(graph[current][i]==1 && visited[i]==0)
    {
      visited[i]=1;
  queue[rear++]=i;
 printf("\nBFS Traced Edge:%d->%d",
      current,i);
```

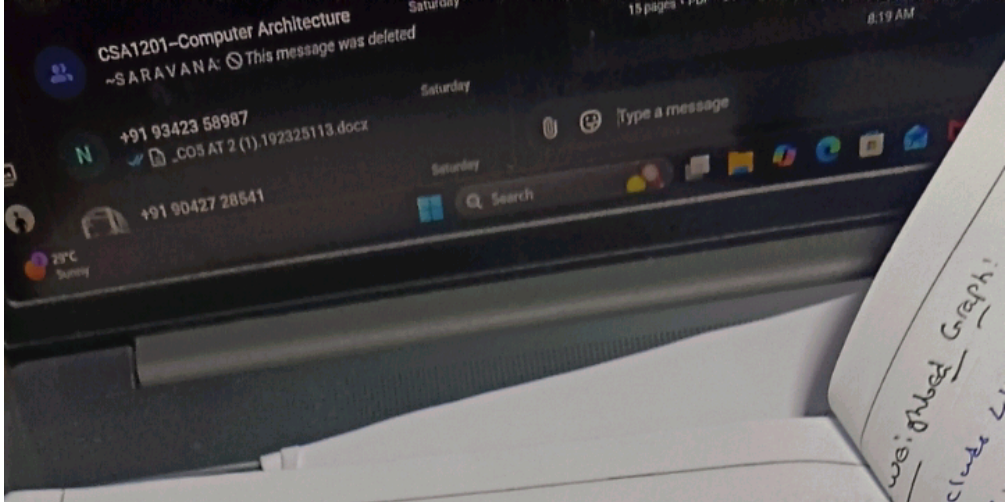
```
    }
  }
}
printf("\n");
}
int main()
{
  int vstart;
  printf("Enter number of vertices");
  scanf("%d", &n);
  printf("Enter adjacency matrix:\n");
  for(int i=0; i<n; i++)
  {
    for(int j=0; j<n; j++)
    { scanf("%d", &graph[i][j]);
    }
  }
    printf("Enter starting vertex:");
  scanf("%d", &vstart);
  BFS(vstart)
  return 0;
}
```

*Output: Enter adjacency matrix:

0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0

Enter starting vertex: 0
BFS Traversal: 0
BFS Tree Edge: 0-1
BFS Tree Edge: 0-2
BFS Tree Edge: 1-3

```
...bra's shortest path Algorithm
...clude <stdio.h>
#include <limits.h>
#define MAX 20
int graph[MAX][MAX];
int distance[MAX];
int visited[MAX];
int n;
int find minvertex()
{
  int min = INT_MAX;
  int minvertex = -1;
  for (int i=0; i<n; i++)
  {
    if (visited[i] == 0 && distance[i] < min)
    {
      min = distance[i];
      minvertex = i;
    }
  }
  return minvertex;
}
void dijkstra's (int source)
{
  for (int i=0; i<n; i++)
  {
    distance[i] = INT_MAX;
    visited[i] = 0;
  }
  distance[source] = 0;
```




```
for (int count=0 ; count <n-1; count++)
{
  int u = FindMinVertex();
  if (u==-1)
    break;
  visited[u]=1;
  for(int v=0; v<n; v++)
  {
    if (!visited[v]&&
       graph[u][v]!=0 &&
      distance[u]!=INT_MAX &&
      distance[u]+graph[u][v]<distance[v])
    {
      distance[v]=
        distance[u]+ graph[u][v];
    }
  }
}
  printf("\n shortest distance from vertex %d:\n",source);
  for(int i=0;i<n;i++)
}
printf("Enter source vertex: ")
scanf("%d",&source)
dijikstra(source);
return 0;
}
```

Output: Shortest distance from vertex 0:

vertex 0 : 0
vertex 1 : 10
vertex 2 : 50
vertex 3 : 30
vertex 4 : 60

```
:onload Graph!
elude <stdio.h>
define MAX 20
nt graph [MAX][MAX];
nt visited [MAX];
nt distance [MAX];
nt parent [MAX];
nt Queue [MAX];
nt n;
nt front=0;
nt rear=0;
void BFS (int source)
{
  for (int i=0;i<n;i++)
  {
    visited [i]=0;
    distance [i]=-1;
    parent [i]=-1;
  }
  visited [source]=1;
  distance [source]=0;
  Queue [rear++]=source;
  while (front<rear)
  {
    int current=Queue [front++]
    for (int i=0;i<n;i++)
    {
      visited [i]=1;
      distance [i]=distance [current]+1;
      parent [i]=current;
      Queue [rear++]=i;
```

```
    }
  }
}
void printPath(int vertex)
{
  if (vertex == -1)
    return;
  printPath(parent[vertex]);
  printf("%d ", vertex);
}
int main()
{
  int source;
  int destination;
  printf("Enter number of vertices: ");
  scanf("%d", &n);
  printf("Enter adjacency matrix:\n");
  for (int i = 0; i < n; i++)
  {
    for (int j = 0; j < n; j++)
    {
      scanf("%d", &graph[i][j]);
    }
  }
  printf("Shortest path: ");
  printPath(destination);
  printf("\n");
  return 0;
}
```

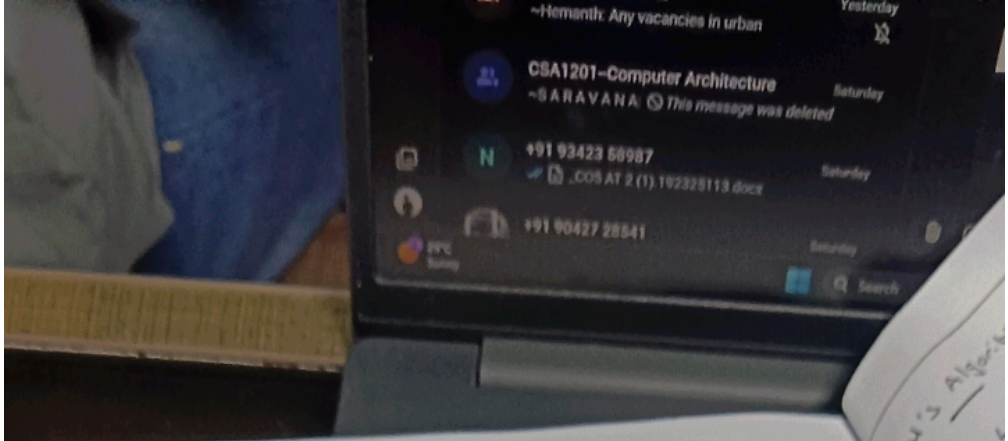
* Output:

Shortest distance = 3

Shortest path = 0 1 3 4

Algorithm

```
#include <stdio.h>
#include <limits.h>
#define MAX 20
int graph[MAX][MAX];
int selected[MAX];
int n;
void prims()
{
  int edge count=0;
  int total weight=0;
selected[0]=1;
printf("\n Edges in minimum spanning Tree:\n");
while (edge count<n-1)
{ int min=INT_MAX;
  int x=-1;
  int y=-1;
  for (int i=0;i<n;i++)
  {
    if (selected[i]==1)
    { for (int j=0;j<n;j++)
      {
        if (selected[j]==0 &&
            graph[i][j]!=0 &&
            graph[i][j]<min)
        { min=graph[i][j];
          x=i;
          y=j;
```




```
    }
  }
 }
}
if (x==-1 || y==-1)
{
  Printf("graph is disconnected\n");
   return;
}
Printf("%d-%d : %d\n", x, y, min);
total weight += min;
selected[y] = 1;
edge count++;
}
Printf("\nTotal MST weight = %d\n", total weight);
}
int main()
{
Printf("Enter number of vertices");
for (i=0; i<n; i++)
{ for (int j=0; j<n; j++)
  { scanf("%d", &graph[i][j]);
  }
}
prims();
return 0;
```

* Output: Enter weighted adjacency matrix for edges in minimum spa

0 2 0 6 0
2 0 7 8 5
0 7 0 0 7
6 8 0 0 9
0 5 7 9 0

0-1 = 2
1-2 = 3
1-4 = 5
0-3 = 6
Total MST weight = 16

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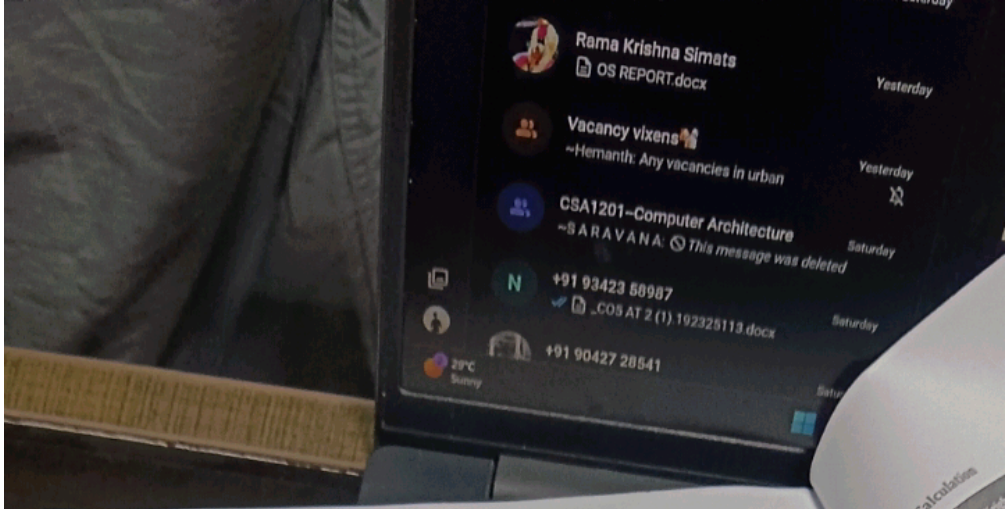
hal's Algorithm:-

```
#include <stdio.h>
#define MAX 100
typedef struct
{
    int source;
    int destination;
    int weight;
} Edge;
Edge Edges[MAX];
int parent[MAX];
int n;
int edge count;
int find(int vertex)
int find(parent[vertex] == vertex)
        return vertex;
return parent[vertex] = find(parent[vertex]);
}
    void union set(int a, int b)
{ int root A = find(a);
  int root B = find(b);
  parent[root B] = root A;
}
    void sort Edges()
    { for (int i = 0; i < edge count-1; i++)
      { for (int j=0; j < edge count-i-1; j++)
        { if (edges[j].weight > edges[j+1].weight
          {
```




```
    }
  }
}
void kruskal()
{ int mstEdge=0;
  int total weight=0;
  sort Edges();
for(int i=0; i<n; i++)
{ parent[i]=i;
}
Print("\n Edges in minimum spanning Tree:\n");
for(int i=0; i<edge count && mstEdge<n-1; i++)
{
  int u=edge[i].source;
  int v=edge[i].destination;
  int root u = find(u);
  int root v = find(v);
  if(root u != root v)
  {
    printf("%d - %d: %d\n", u, v, edge[i].weight);
  }
  kruskal();
  return 0;
}
```

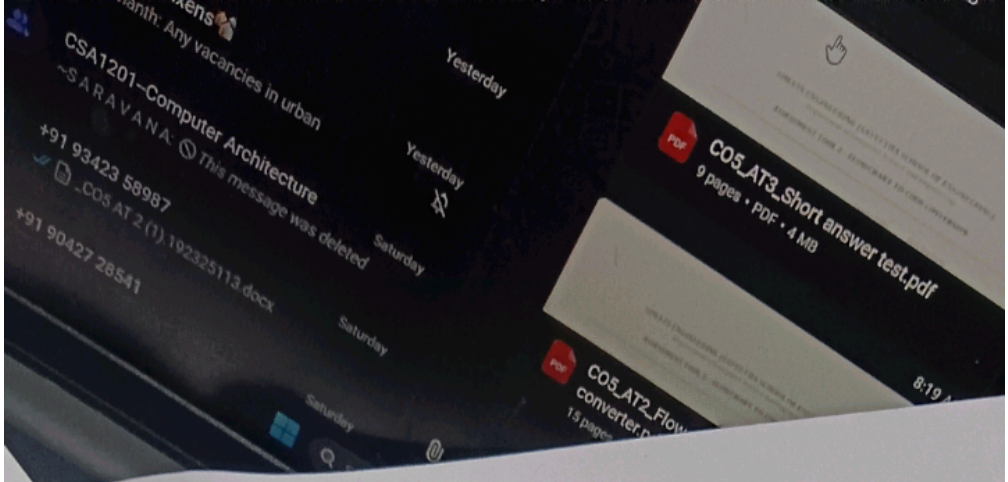
*Output: Enter edges:

0 1 2
0 3 6
1 2 3
1 3 8
1 4 5
2 4 7
3 4 9

Edges in Minimum spanning Tree:

0-1 : 2
1-2 : 3
1-4 : 5
0-3 : 6

Total MST weight = 16



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date:	Date:	Date: